Peg solitaire on unicyclic graphs: crowns, cycle-critical graphs, and two atlases

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Abstract

We study peg solitaire on graphs in the sense of Beeler and Hoilman. Our starting point is a pair of exhaustive computational atlases: the solvability status, for every starting hole, of all 205,001 trees on up to 18 vertices and all 61,130 unicyclic graphs on up to 14 vertices. Prior exhaustive data stopped at all graphs on 7 vertices and freely solvable trees on 10. The atlases lead to several theorems and one new phenomenon. For the *crown* $C(k;p)$, a cycle with p pendant vertices at a single vertex, we determine the solvability threshold $p^*(k)$ exactly for all $k \leq 31$ and prove that the capacity of a cycle is bounded by an absolute constant: $C(k;p)$ is unsolvable for every k once $p \geq 13$. The sufficiency half is proved for all k via a two-jump purge gadget and a new family of single-cluster caterpillar lemmas, which also yield a position law: a cluster of p pendants is tolerated on a long path only at spine position $2p-1$ (even paths) or $2(p-1)$ (odd paths). We then isolate the graphs responsible for the failure of a natural reduction: call a solvable unicyclic graph *cycle-critical* if every spanning tree is unsolvable. There are 1,839 such graphs on at most 14 vertices, they come in exactly two tiers, and they include an infinite two-parameter family, the double crowns $DC(k,d)$, for which we prove a parity dichotomy: $DC(k,d)$ with two pendant pairs at distance d is freely reachable from its spanning trees when d is odd, and cycle-critical when d is even (solvability proved for all $k \equiv 0 \pmod 4$; criticality reduced to, and following from, a caterpillar lemma verified at all positions through spine length 20). Finally we prove that all interaction in a crown game is bounded by constants independent of k, measure the Myhill–Nerode complexity of the reachable configuration spaces, and explain why the eventual periodicity of $k \mapsto$ "$C(k;p)$ is solvable", which all our data supports, resists the standard regularity toolkit. We also correct a boundary case of a published theorem on windmill graphs. All computations are exact and reproducible from the accompanying repository.

1 Introduction

Peg solitaire on graphs was introduced by Beeler and Hoilman [1]. Pegs occupy all vertices of a connected graph G except one, the starting hole; a move takes a path $u\,v\,w$ with pegs on u, v and a hole on w, jumps the peg from u to w, and removes the peg on v. The graph is *solvable* if from some starting hole a sequence of jumps leaves a single peg, and *freely solvable* if this works from every hole. The central open problem of the area, posed in [1] and repeated in the survey [7], is to characterize the solvable trees; progress has come one family at a time, through double stars [2], trees of diameter four [4], caterpillars [5], and banana trees [6], among others.

This paper approaches the subject from the unicyclic side, where, to our knowledge, no systematic study existed. Two decisions shaped the work. First, we computed before we conjectured: every claim below grew out of an exhaustive, exactly verified dataset. Second, we kept the boundary between what is proved, what is machine-verified (and therefore proved, when the claim is finite), and what is conjectured, as sharp as we could make it.

Our contributions:

1. **Two atlases** (Section 3). The solvability classification of all trees on $4 \leq n \leq 18$ vertices (205,001 trees; the prior frontier was $n \leq 10$, and only for free solvability [3]) and of all unicyclic graphs on $n \leq 14$ vertices (61,130 graphs). About 61% of large trees are solvable and about 24% freely solvable; unicyclic graphs are far more permissive, at roughly 85% solvable. As a first dividend the atlas caught a boundary error in a published theorem: the windmill variant $W(1,1)$ of [2] is claimed freely solvable and is not (Proposition 2.3).

2. **Single-cluster caterpillars and the position law** (Section 4). We compute the pendant capacity $\mathrm{cap}(P_m, i)$, the largest cluster of pendants attachable at spine position i of an m-vertex path without destroying solvability, for all $m \leq 20$. Even paths have capacity zero at depth five and beyond; odd paths obey an eventually periodic profile peaking at position six. The solvable stable positions follow the law $i = 2p - 1$ (even m) and $i = 2(p - 1)$ (odd m). A two-jump *purge gadget* turns machine-found base cases into theorems for all lengths (Theorem 4.3).

3. **Crowns** (Section 5). Write $C(k; p)$ for a k-cycle with p pendants at one vertex. We determine the threshold $p^*(k)$, below which $C(k; p)$ is solvable and above which it is not, exactly for all $k \leq 31$ (Theorem 5.1): $p^* = 4$ for odd $13 \leq k \leq 31$, $p^* = 2$ for even $10 \leq k \leq 30$, with ten exceptional small values. The sufficiency half holds for all k by inheritance from the caterpillar families. For necessity we prove a golden-ratio weight bound (Theorem 5.3): *no* cycle, of any length, tolerates 13 pendants at a vertex. En route we prove that a fully pegged path can deliver at most one peg to an adjacent drained cell, ever (Lemma 5.4), and that all interface activity in a crown game is bounded by constants independent of k (Section 8).

4. **Cycle-critical graphs** (Section 6). A solvable unicyclic graph is cycle-critical if every spanning tree is unsolvable: the cycle edge is load-bearing. The atlas contains 1,839 examples on $n \leq 14$ vertices; the smallest is C_4 with one pendant on each of two opposite vertices. Exactly two tiers occur: for 1,810 graphs the best spanning tree strands two pegs, for the remaining 29 it strands three. We conjecture this is sharp in general: one extra edge rescues at most two pegs (Conjecture 6.3).

5. **Double crowns and a parity dichotomy** (Section 7). $DC(k, d)$ is a k-cycle with two pendants at x_0 and two at x_d. For odd d we prove $DC(k, d)$ solvable together with its spanning trees (Theorem 7.1); for even d it is cycle-critical in every case we can test, and we prove the solvability half for the antipodal family for all $k \equiv 0 \pmod 4$, $k \geq 16$, by an explicit circulating play family (Theorem 7.2). The solvable starting holes obey a clean law: they sit at distances $\{3, 6, k/2 - 6, k/2 - 3\}$ from the clusters (Proposition 7.6).

6. **The periodicity question** (Section 8). All data says $k \mapsto p^*(k)$ is eventually periodic with period two. We prove the three bounded-interface lemmas that any closure argument will need, and then measure why the obvious closure fails: the reachable configuration languages of crown games have super-linearly growing Myhill–Nerode complexity under the natural encodings, so no k-uniform regular invariant exists. The question joins the

octal-games periodicity problem [9] in the class of statements that are almost certainly true, cheap to verify instance-by-instance, and apparently expensive to close.

Throughout, “machine-verified” means an exhaustive, exact search whose code and outputs are in the public repository accompanying this paper. Finite claims verified this way are proved; we reserve “conjecture” for statements quantified over infinitely many untested cases. Two of our own intermediate models contained errors that were caught by cross-implementation disagreement and by testing consequences against known instances; we record the corrected statements only, but the reader should know the pipeline was adversarial toward itself.

2 Preliminaries

Graphs are finite, simple, connected. P_n and C_n denote the path and cycle on n vertices. A *pendant* is a vertex of degree one. For a graph G with starting hole s, a *terminal state* is one admitting no jump; G is solvable from s if some play from s terminates with one peg. We write $\mathrm{ps}(G)$ for the minimum number of pegs over all terminal states and all starting holes, so G is solvable iff $\mathrm{ps}(G) = 1$.

We use three known results freely.

Proposition 2.1 ([1, 2])**.** *P_n is solvable iff n is even or $n = 3$, and freely solvable iff $n = 2$. C_n is solvable iff n is even or $n = 3$. The double star $DS(L, R)$, $L \geq R$, is solvable iff $L \leq R + 1$.*

Proposition 2.2 (Inheritance [5])**.** *If a spanning subgraph of G is solvable from hole s, so is G.*

Our solver performs exact memoized search over peg configurations, recording for every starting hole the minimum and maximum terminal peg counts. It was validated against every exactly-stated published theorem we could encode: 122 instances covering paths, cycles, complete graphs, stars, double stars and windmills, all of which agree, with one exception that turned out to be an error in the literature rather than in the solver.

Proposition 2.3. *Theorem 2.2(ii) of [2] states that the windmill variant $W(P, B)$, a universal vertex u carrying B triangles and P pendants, is freely solvable iff $P \leq 2B-1$ and $(P, B) \neq (0, 2)$. This fails at $(P, B) = (1, 1)$: $W(1, 1)$ is solvable but not freely solvable.*

Proof. Start the hole at the pendant p_1. Every first jump must land in p_1 and jump over u, since u is the only neighbour of p_1. Either blade vertex may jump; both leave the two remaining pegs on b_i and p_1, which are non-adjacent, so the game ends with two pegs. The published sufficiency proof checks holes on u and on blades but not on pendants. (Solvability from blade holes is easily checked, so the graph is solvable.) □

3 The atlases

Trees were generated by the Wright–Richmond–Odlyzko–McKay algorithm; unicyclic graphs as necklaces of rooted trees under the dihedral action, validated in two independent ways (counts match OEIS A001429, and a Burnside/cycle-index computation matches per cycle length). Every graph was solved exactly for every starting hole.

Proposition 3.1. *Of the 205,001 trees with $4 \leq n \leq 18$: at $n = 18$, 61.0% are solvable and 23.6% freely solvable. There are no freely solvable trees at $n \in \{4, 5, 7\}$. Of the 61,130 unicyclic graphs with $4 \leq n \leq 14$: at $n = 14$, 84.7% are solvable and more than half are freely solvable.*

The contrast between 61% and 85% is the paper in one line: a single extra edge changes the game, and the rest of the paper is about how.

4 Single-cluster caterpillars

Write $P_m(i:p)$ for the caterpillar consisting of a spine $x_0 \cdots x_{m-1}$ with p pendants at x_i, and define the *pendant capacity* $\operatorname{cap}(P_m, i)$ as the largest p for which $P_m(i:p)$ is solvable (from some hole), or 0 if none. Beeler, Green and Harper [5] studied caterpillars with clusters at the spine ends; interior single clusters seem not to have been treated.

Proposition 4.1 (machine-verified, $m \leq 20$)**.** *For even $m \geq 14$ the capacity profile in $i = 0, 1, 2, \ldots$ is $0, 1, 1, 2, 2, 0, 0, \ldots$: in particular a single pendant at depth ≥ 5 makes an even path unsolvable. For odd $m \geq 13$ the profile is $2, 0, 2, 0, 3, 0, 4, 0, 2, 0, 2, \ldots$: zero at odd depths, peaking at $i = 6$, settling to 2. Small m carry exceptions, e.g.* $\operatorname{cap}(P_{11}, 4) = 5$ *and* $\operatorname{cap}(P_9, 2) = \operatorname{cap}(P_9, 4) = 4$.

The stable solvable positions obey a law: for a cluster of size p, position $2p-1$ on even spines and $2(p-1)$ on odd spines. Each pendant needs two spine vertices of runway. The following gadget turns the finite data into theorems.

Lemma 4.2 (Purge gadget)**.** *Let H contain an induced path $y\, x_{k-2}\, x_{k-1}\, x_k\, x_{k+1}$ in which x_{k-1}, x_k, x_{k+1} have no further neighbours. From the full configuration with hole at x_k, the two jumps $x_{k-2} > x_{k-1} > x_k$ and $x_{k+1} > x_k > x_{k-1}$ leave x_k, x_{k+1} permanently empty and the graph $H - \{x_k, x_{k+1}\}$ fully pegged with hole at x_{k-2}. Hence if $H - \{x_k, x_{k+1}\}$ is solvable from x_{k-2}, then H is solvable from x_k.*

Proof. Legality of the two jumps is immediate. Afterwards no path of length two reaches x_k or x_{k+1} through a pegged middle, and any play of $H - \{x_k, x_{k+1}\}$ is a play of H verbatim, since the two dead vertices never regain pegs. □

Theorem 4.3. *For all spine lengths of the appropriate parity, $P_k(1:1)$ and $P_k(3:2)$ are solvable for even $k \geq 6$; $P_k(0:1)$ and $P_k(2:2)$ for odd $k \geq 7$; $P_k(4:3)$ for odd $k \geq 9$; and $P_k(6:4)$ for odd $k \geq 13$. In each case the starting hole may be taken at x_{k-2}.*

Proof. Induction on k in steps of two. The base cases are the following explicit plays, each verified mechanically (spine vertices are numbered $0, \ldots, k-1$, pendants from k upward):

$P_6(1:1)$	2>3>4 0>1>2 5>4>3 3>2>1 6>1>0
$P_6(3:2)$	6>3>4 1>2>3 7>3>2 5>4>3 3>2>1 0>1>2
$P_7(0:1)$	3>4>5 1>2>3 7>0>1 6>5>4 4>3>2 2>1>0
$P_7(2:2)$	3>4>5 1>2>3 6>5>4 4>3>2 7>2>1 0>1>2 8>2>1
$P_9(4:3)$	5>6>7 3>4>5 8>7>6 6>5>4 9>4>3 2>3>4 0>1>2 10>4>3 2>3>4 11>4>3
$P_{13}(6:4)$	9>10>11 7>8>9 13>6>7 4>5>6 2>3>4 0>1>2 14>6>5 5>4>3 2>3>4 12>11>10 10>9>8 8>7>6 15>6>5 4>5>6 16>6>5

For the step, $P_{k+2}(i:p)$ satisfies the hypothesis of Lemma 4.2 at its far end: the three outermost spine vertices carry no pendants because $i \leq 6 < k-2$. The gadget reduces the full board with hole x_k to $P_k(i:p)$ with hole x_{k-2}, solvable by induction. □

Remark 4.4. Machine search shows these positions are the only stable ones: for $p = 2$ on long even spines the solvable positions are exactly $i \in \{3, 4\}$, and for $p = 4$ on long odd spines exactly $i = 6$. Single-cluster caterpillars cap at $p = 2$ (even spine) and $p = 4$ (odd), which the next section shows are precisely the crown capacities.

5 Crowns

Let $C(k; p)$ be the cycle $x_0 x_1 \cdots x_{k-1}$ with $p \geq 1$ pendants at x_0.

Theorem 5.1. *For all $3 \leq k \leq 31$, $C(k; p)$ is solvable iff $p \leq p^*(k)$, where $p^*(k) = 4$ for odd $k \geq 13$, $p^*(k) = 2$ for even $k \geq 10$, and $(p^*(3), \ldots, p^*(12)) = (2, 1, 3, 2, 3, 3, 4, 2, 5, 2)$. Moreover the solvable range is an interval in p, and, for every k in this range, $p^*(k) = \max_i \operatorname{cap}(P_k, i)$: deleting the right cycle edge never costs anything.*

Proof. Sufficiency for large k is Theorem 4.3 plus Proposition 2.2: delete the cycle edge at the stable distance from x_0 and inherit from the caterpillar. Exceptional rows and small cases are finite and machine-witnessed. Necessity for $p^* < p \leq 13$ was verified exhaustively, from every starting hole, for even $k \leq 30$ and odd $k \leq 31$ (211 instances), using a pendant-symmetric solver (pendants are interchangeable, so a configuration is a cycle mask plus a counter) cross-validated against the direct solver. Necessity for $p \geq 13$ and *all* k is Theorem 5.3 below. □

Two remarks before the uniform bound. First, the theorem has a counterintuitive corollary worth isolating.

Corollary 5.2. *Bare odd cycles C_k, $k \geq 5$, are unsolvable, yet $C(k; p)$ is solvable for $1 \leq p \leq p^*(k)$: attaching a single pendant rescues any odd cycle.*

Second, for fixed p the map $k \mapsto [\, p \leq p^*(k) \,]$ appears eventually periodic with period two; Section 8 discusses why we do not know how to close this and state it here only for $k \leq 31$.

5.1 A uniform capacity bound

Let φ be the golden ratio and $s = 1/\varphi$, so $s^2 + s = 1$.

Theorem 5.3. *For every $k \geq 3$ and every $p \geq 13$, $C(k; p)$ is unsolvable.*

Proof. Weight the cycle by $w(x_j) = s^{\min(j, k-j)}$ and give each pendant weight $t \in (0, s^2)$. Let W be the total weight of pegged vertices. A jump along $u\,v\,w$ changes W by $w(w) - w(u) - w(v)$. Jumps toward x_0 along either arc change W by $s^d - s^{d+1} - s^{d+2} = s^d(1 - s - s^2) = 0$; jumps away, and the finitely many antipodal orientations, are strictly negative (e.g. the antipodal even case gives $-s^{k/2}$). Refills $x_2 > x_1 > x_0$ change W by $1 - s - s^2 = 0$. Classify the jumps that consume the peg on x_0: a pendant exiting over x_0 burns $1 + t - s > 1 - s = s^2$; a pendant hopping to an empty pendant slot burns 1; a cycle neighbour feeding a pendant burns $1 + s - t > 0$; x_0 jumping into the cycle burns $1 + s - s^2 > 0$; a through jump $x_1 > x_0 > x_{k-1}$ burns 1. So no jump increases W, and each pendant-exit event burns more than s^2.

Pendants have degree one, so a pendant peg leaves the cluster only by exiting over x_0, and pendant-to-pendant hops do not change the number of pegged pendants. If a pendant-exits and c feeds occur and $\varepsilon \in \{0, 1\}$ pendant pegs remain at the end, then $a - c = p - \varepsilon$, so $a \geq p - 1$. The initial weight satisfies $W_0 < 1 + pt + 2\sum_{d \geq 1} s^d = 1 + pt + 2\varphi$, and W remains positive, so $s^2(p - 1) \leq s^2 a < 1 + 2\varphi + pt$. Letting $t \to 0$ and using $1/s^2 = \varphi + 1$,

$$p - 1 \;\leq\; (1 + 2\varphi)(\varphi + 1) \;=\; 5\varphi + 3 \;=\; 11.09\ldots,$$

whence $p \leq 12$. □

The constant is far from the truth ($p^* \leq 5$ everywhere), but it is uniform in k, which is what the structure of Theorem 5.1 needs: the sharp threshold is then a statement about the finite strip $p^* < p \leq 12$ only.

5.2 One-shot delivery

The mechanism limiting p^* deserves its own statement. Consider a path $1, 2, \dots, m$, fully pegged, adjacent at cell 1 to an external cell 0 that an adversary may drain at any time; a *delivery* is a jump $2 > 1 > 0$.

Lemma 5.4 (One-shot delivery)**.** *A fully pegged path delivers at most one peg, regardless of length: after the first delivery no second delivery is possible. With drains at both ends, at most one delivery per end.*

Proof. After a delivery, cells $1, 2$ are empty. We claim that once cells $j, j+1$ are empty and all cells left of j are empty, cell j can never again hold a peg; the lemma follows with $j = 1$. A peg can arrive at j only by the jump $j{+}2 > j{+}1 > j$, which requires a peg at $j + 1$; a peg can arrive at $j + 1$ only from $j{+}3 > j{+}2 > j{+}1$, which empties $j + 2$ and $j + 3$ and recreates the hypothesis two cells to the right. The induction is well-founded because the path is finite. □

In a crown, all refills of x_0 beyond the two supplied by the arc must therefore be financed by pendant exits landing on the gate cells: a circular economy in which each refill costs a refill. The exact bookkeeping is delicate (Section 8), but the following consequence of the same weighting technique is not.

Lemma 5.5. *In any solving play of any $C(k; p)$, the number r of jumps landing on x_0 satisfies $r \le 10$; consequently the total number of jumps involving x_0 or a pendant is at most 21.*

Proof. Let Φ be the pegged weight of the arc $x_1, \dots, x_{k-1}$ under $w(x_j) = s^{\min(j,k-j)}$. Refills remove $s + s^2 = 1$ from Φ; pendant exits add s; through jumps are neutral; all other jumps do not increase Φ. Hence $r \le \Phi_0 + sa \le 2\varphi + sa$. Every jump involving x_0 or a pendant consumes the peg on x_0, whose supply is at most $1 + r$, so $a \le 1 + r$ and $r(1 - s) \le 2\varphi + s$, giving $r \le (2\varphi + s)/s^2 = 10.09\dots$ □

6 Cycle-critical graphs

The equality $p^*(k) = \max_i \operatorname{cap}(P_k, i)$ of Theorem 5.1 suggests a general principle: perhaps a unicyclic graph is solvable iff some spanning tree is. The atlas refutes this decisively.

Definition 6.1. A solvable unicyclic graph G is *cycle-critical* if $G - e$ is unsolvable for every cycle edge e, i.e. every spanning tree of G is unsolvable.

Proposition 6.2 (machine-verified)**.** *Exactly* 1,839 *of the* 52,021 *solvable unicyclic graphs on $n \le 14$ vertices are cycle-critical, with counts $1, 6, 9, 12, 91, 126, 398, 1196$ for $n = 6, \dots, 14$. The smallest is C_4 with one pendant on each of two opposite vertices. For* 1,810 *of them the best spanning tree strands exactly two pegs; for the remaining* 29*, every spanning tree strands at least three. No solvable unicyclic graph on $n \le 14$ vertices has all spanning trees stranding four or more.*

Since a play that never uses a jump path through e is verbatim a play of $G - e$, criticality says every solution *circulates*: it uses every cycle edge. We tested two structural explanations and both fail: the two stranded pegs of the best spanning tree rarely sit at the endpoints of the deleted edge (5 cases of 28 at $n \le 10$), and there are critical graphs all of whose solutions have zero net flow around the cycle (gross usage does not cancel; net usage can). Criticality is a statement about gross traversal, and its characterization is open. The data supports a clean bound.

Conjecture 6.3 (Rescue bound)**.** *For every unicyclic G, writing $R(G) = \min_T \mathrm{ps}(T) - \mathrm{ps}(G)$ over spanning trees T, one has $R(G) \le 2$: one extra edge rescues at most two pegs.*

This is machine-verified for every unicyclic graph on $n \le 12$ vertices, solvable or not: $R = 0$ in 7437 cases, $R = 1$ in 421, and $R = 2$ in exactly 13, so the bound is both true and tight. The extremal cases are essentially unique in form: twelve of the thirteen are triangles carrying a heavy pendant cluster at one corner, and the last is a C_8 with two pendant pairs. The triangle is K_3, whose third edge lets the loaded corner be refilled across the clique, worth two pegs when a large cluster is cleared; this is the natural candidate for a worst case.

One direction is provable and pins down where the difficulty lies. Fix an optimal play P of G (ending with $r = \mathrm{ps}(G)$ pegs) and a cycle edge e. Replay P greedily on $T = G - e$: process its jumps in order, keeping each that is legal in the current T-state and skipping the rest.

Lemma 6.4 (Skip identity)**.** *The greedy replay ends with exactly $r + s_e$ pegs, where s_e is the number of jumps it skips; and $s_e = 0$ if P never traverses e. Hence $R(G) \le \min_e s_e$, and $R(G) \le 0$ whenever some optimal play of G omits a cycle edge.*

Proof. Both G under P and T under the replay lose exactly one peg per executed jump; G executes all N jumps to reach r pegs from $n-1$, and T executes $N - s_e$, reaching $(n-1) - (N - s_e) = r + s_e$. Before the first skip the two states agree, so the first skipped jump must be one that uses e; if none does, there are no skips. □

In particular a positive rescue forces every optimal play to traverse every cycle edge — for solvable G, exactly cycle-criticality. This does *not* prove $R \le 2$, and instructively so: the greedy replay of a *fixed* optimal play can cascade, and we have measured $\min_e s_e$ as large as 6 while the true R stayed at most 2. The tree recovers the lost pegs with a play of its own that the inherited one does not see. So the honest state of the bound is this: verified and tight through $n = 12$, reduced by Lemma 6.4 to controlling skips, but not closed, because inheritance from a single play is too lossy and the tree's native optimum must be used instead. We regard it as the most accessible of the paper's open problems, but it is open.

The mechanism behind criticality and the caterpillar necessity is the same, and can be stated cleanly.

Lemma 6.5 (Refill necessity)**.** *Let v have exactly two pendant neighbours p_1, p_2. Any play that removes the pegs on both p_1 and p_2 contains a jump landing on v, occurring after a moment when v is empty.*

Proof. A leaf p_r is removed only as the jumper of a jump over its unique neighbour v (it cannot be a middle, and it is never created). Such a jump needs v pegged immediately before and leaves v empty immediately after. Let J_1, then J_2, be the jumps removing p_1, p_2. After J_1 the vertex v is empty; just before J_2 it is pegged; so some intervening jump lands on v, its middle a non-pendant neighbour of v. □

In a double-cluster caterpillar this forces both x_i and x_j to be refilled from the spine. The inter-cluster segment has $j - i - 1$ vertices, odd exactly when the separation $j - i$ is even; that the two refills and the spine clearance cannot be reconciled on an odd interior segment is what Proposition 7.3 asserts and what we cannot yet prove — the same obstruction, on a path, as the crown refill ledger.

7 Double crowns

Let $DC(k,d)$ be the k-cycle with two pendants at x_0 and two at x_d, $2 \le d \le k/2$. This family witnesses cycle-criticality infinitely often, and the deciding parameter is the parity of d.

Theorem 7.1 (Odd separation)**.** *For $d \in \{3,5,7\}$ and every even $k \ge 2d+2$, the caterpillar obtained from $DC(k,d)$ by deleting the cycle edge $x_{k-1}x_0$, namely the double-cluster caterpillar with pairs at spine positions 0 and d, is solvable with hole at x_{k-2}. Consequently $DC(k,d)$ is solvable and not cycle-critical.*

Proof. The far spine end is three pendant-free vertices, so Lemma 4.2 applies and reduces $k+2$ to k exactly as in Theorem 4.3; the base cases at $k = 2d+2$ are machine-witnessed. The same scheme works for any fixed odd d; we state the cases whose base plays we have exhibited. Machine search shows, additionally, that the cluster pair may sit at any even offset i with the same conclusion, and at no odd offset. □

Theorem 7.2 (Even separation: solvability)**.** *For every $k \equiv 0 \pmod 4$ with $k \ge 16$, the antipodal double crown $DC(k,k/2)$ is solvable with starting hole x_3.*

Proof. Write $d = k/2$ and let the pendants at x_0 be q_1, q_2 and at x_d be q_3, q_4. We exhibit the play in six phases and prove each legal; cells not mentioned retain their state.

Phase A: 5>4>3, 2>3>4, 0>1>2. Directly legal from the start (hole x_3); afterwards the holes are exactly $\{0,1,3,5\}$.

Phase B: for $j = 0,1,\ldots,(d-8)/2$, the pair $(2j+7) > (2j+6) > (2j+5)$ then $(2j+4) > (2j+5) > (2j+6)$. We claim the invariant: after pair j the holes are $\{0,1\} \cup \{3,\ldots,2j+5\} \cup \{2j+7\}$, with pegs on 2, on $2j+6$, and on everything from $2j+8$ onward. For $j = 0$ this is a direct check from the state after A. Given the invariant at j, pair $j+1$ is legal: cells $2j+9, 2j+8$ are pegged and $2j+7$ is a hole for the first jump, which fills $2j+7$ and empties $2j+8, 2j+9$; then $2j+6$ (pegged by the invariant) jumps over the just-filled $2j+7$ into the just-emptied $2j+8$. The new state satisfies the invariant at $j+1$. After the last pair the holes are $\{0,1\} \cup \{3,\ldots,d-3\} \cup \{d-1\}$ and the pegs on the low side are exactly x_2 and x_{d-2}; cells $x_d,\ldots,x_{k-1}$ are untouched.

Phase C: $(k-2)$>$(k-1)$>0, then m>$(m+1)$>$(m+2)$ for $m = k-4, k-6, \ldots, d+4$. The first jump is legal (x_0 is a hole from A) and empties x_{k-2}, x_{k-1}. Each subsequent step uses two untouched pegged cells $m, m+1$ and the hole at $m+2$ left by the previous step; it fills $m+2$ and empties $m, m+1$. After C the pegged cells above x_d are exactly $x_{d+1}, x_{d+2}, x_{d+3}$ (never touched) and $x_{d+6}, x_{d+8}, \ldots, x_{k-2}$, together with x_0.

Phase D: q>0>1 with $q = q_1$, then 2>1>0, then q>0>$(k-1)$ with $q = q_2$. Legal in turn: x_0 pegged and x_1 empty; then x_2, x_1 pegged, x_0 empty; then x_0 pegged again and x_{k-1} empty from C. The cluster at x_0 is now gone and x_{k-1} is pegged.

Phase E: m>$(m-1)$>$(m-2)$ for $m = k-1, k-3, \ldots, d+7$. Step m uses the peg at m (from D when $m = k-1$, else from the previous step), the peg at $m-1$ (these are the cells $x_{d+6},\ldots,x_{k-2}$ left by C), and the hole at $m-2$ (emptied by C). After E the only pegs above x_d are $x_{d+1}, x_{d+2}, x_{d+3}$ and x_{d+5}.

Phase F: q>d>$(d-1)$, $(d+2)$>$(d+1)$>d, q>d>$(d+1)$, $(d-2)$>$(d-1)$>d, d>$(d+1)$>$(d+2)$, $(d+2)$>$(d+3)$>$(d+4)$, $(d+5)$>$(d+4)$>$(d+3)$. Entering F the pegs are exactly $x_{d-2}, x_d, x_{d+1}, x_{d+2}, x_{d+3}, x_{d+5}$ plus the two pendants at x_d; each jump is checked directly against the previous line, and the final state is a single peg on x_{d+3}.

The phase lengths sum to $k+2$, which equals the number of vertices minus two, as they must. For $k = 16$ phases B, C, E consist of one pair, two jumps, and one jump respectively, and the general description specializes correctly; the play has also been verified mechanically for $k = 16, 20, 24, 28, 32, 36, 40$. □

The criticality half follows from a caterpillar necessity statement and a clean reduction. Write $P_m(2@i, 2@j)$ for the path on m vertices with two pendants at x_i and two at x_j.

Proposition 7.3 (Caterpillar necessity; machine-verified)**.** *Let m be even, $14 \leq m \leq 20$. For every $i < j$ with $j - i$ even, $P_m(2@i, 2@j)$ is unsolvable, at all positions. The parity of m matters: for odd m even-separation solvable pairs occur (e.g. several at $m = 19$). We conjecture the statement for all even m.*

Lemma 7.4 (Reduction)**.** *For even k, every spanning tree of $DC(k, d)$ is a double-cluster caterpillar of spine length k whose two pairs are separated by d or $k - d$.*

Proof. Deleting a cycle edge turns the k-cycle into a path on its k vertices carrying the two pairs; the loaded vertices are joined the short way (separation d) or the long way (separation $k - d$) according to which arc holds the deleted edge. □

Theorem 7.5 (Even separation: criticality)**.** *Let k, d be even. Then d, $k - d$ and the spine length k are all even, so by Lemma 7.4 every spanning tree of $DC(k, d)$ is an even-spine, even-separation double-cluster caterpillar. Granting Proposition 7.3 at spine length k, all spanning trees are unsolvable, so $DC(k, d)$, once solvable, is cycle-critical. This is unconditional for even $k \leq 20$, and verified directly through $k = 24$ for $(k, d) \in \{(8, 4), (12, 4), (12, 6), (16, 8), (20, 4), (20, 10), (22, 6), (22, 10), (24, 8), (2$ For d odd, $DC(k, d)$ is freely solvable with solvable spanning trees in every tested case.*

The dichotomy is therefore a theorem in a precise sense: the odd case is proved (Theorem 7.1); the even case is proved modulo extending the finite Proposition 7.3 to all even m, a k-uniform necessity of the same character as the crown strip (Section 8), together with solvability, which is unconditional for the antipodal family (Theorem 7.2) and machine-verified otherwise. The solvable holes carry unexpected structure:

Proposition 7.6 (machine-verified, $k \leq 32$)**.** *For $k \equiv 0 \pmod 4$, $16 \leq k \leq 32$, the solvable starting holes of $DC(k, k/2)$ are exactly the cycle vertices at distance 3, 6, $k/2 - 6$ or $k/2 - 3$ from one of the two loaded vertices. ($DC(12, 6)$ is freely solvable; $DC(8, 4)$ has its own small pattern.)*

The recurring 3 and 6 are the caterpillar runway constants of Section 4, which we do not believe is a coincidence, though we cannot yet derive one law from the other.

8 Bounded interfaces and the periodicity question

Everything we know says the following is true.

Conjecture 8.1. *For each fixed p, the set of k with $C(k; p)$ solvable is eventually periodic in k (with period 2 and threshold 13, on the evidence).*

This would extend Theorem 5.1 from $k \leq 31$ to all k. We record what we can prove toward it, and then quantify the obstruction.

First, all interaction in a crown game is uniformly bounded. Jumps landing on x_0 number at most 10 and jumps meeting the pendant cluster at most 21 (Lemma 5.5); and, weighting distances from the cell antipodal to x_0 instead, the same technique bounds the number of jumps whose path crosses the antipodal cut by an absolute constant (25 with our loose accounting). So for large k a solving play decomposes into $O(1)$ interface events separated by episodes of pure one-dimensional solitaire on segments. One-dimensional solitaire has a rational reachability theory in the sense of Moore and Eppstein [8], and one would like to conclude regularity, hence eventual periodicity, by composition.

The obstruction is that our segments are not free-standing: they begin fully pegged and are driven at their boundary by the bounded event stream. We tested the natural closure route, regular model checking: exhibit a regular invariant containing the reachable configurations and excluding near-solved ones, and verify closure under the move relation symbolically. That route is quantitatively impossible here. Computing the exact reachable configuration sets of $C(k;3)$ from all starting holes and measuring Myhill–Nerode residual counts (lower bounds on any recognizing DFA):

k	10	12	14	16	18
linear encoding	41	93	186	308	—
interleaved encoding	43	97	197	346	479

The counts grow super-linearly under both the around-the-cycle encoding and the two-arms-from-the-gadget encoding: the reachable languages are not k-uniformly regular, so no small regular invariant exists in these encodings. This does not contradict [8], whose regularity concerns undriven paths, and it does not make Conjecture 8.1 doubtful; it locates the difficulty. Either a different abstraction tames driven segments, or the periodicity needs a direct combinatorial argument. The situation is reminiscent of the octal-games periodicity conjecture [9]: overwhelming instance evidence, bounded-looking behaviour, no closure.

The caterpillar necessity of Proposition 7.3 is a twin of this question, and behaves identically. Its reachable configuration sets, measured the same way for the even-separation family as the spine grows, have Myhill–Nerode residual counts $49, 94, 233, 296, 497$ at $m = 8, 10, 12, 14, 16$: again super-linear, again non-regular. We also checked the two most natural certificates directly and both fail: the bipartite two-colouring gives a resource count with no separation-parity constraint, and the exact modular invariants $\sum_{\text{pegs}} w \bmod q$ collapse to the trivial weighting, because at a vertex carrying two pendants the three cherries $p_1 x_i p_2$, $p_1 x_i x_{i\pm 1}$ force $2w(x_i) \equiv 0$ and equal neighbour weights, which propagate along the Fibonacci spine recurrence to kill every nonzero solution. So both of our k-uniform necessity statements — the crown strip and the even-separation caterpillar — are hard for the same structural reason, and neither yields to the standard invariant toolkit. We regard closing either as the main problem left open by this work.

9 Open problems

1. Prove Conjecture 8.1, even for $p = 3$.

2. Characterize the cycle-critical unicyclic graphs; failing that, prove the Rescue Bound (Conjecture 6.3), or its natural extension $\min_T \mathrm{ps}(T) \le \mathrm{ps}(G) + 2c$ for graphs with cycle rank c.

3. Extend Proposition 7.3 to all even spine lengths: an even-spine double-cluster caterpillar with even separation is unsolvable. By Theorem 7.5 this closes the criticality half of

the double crown dichotomy. The obstruction is not the bipartite two-colouring resource count, which we checked gives no separation-parity constraint; a pagoda-type function is the natural candidate.

4. Explain the Distance-$\{3, 6\}$ law (Proposition 7.6) from the runway law of Section 4.

5. Characterize solvable double-cluster caterpillars $P_m(2@i,\, 2@j)$; our data suggests parity of $j - i$ dominates, mirroring Theorem 7.1.

6. The 29 tier-two critical graphs (all spanning trees strand three pegs) await any structural explanation; 23 of the 29 are triangles carrying one heavy cluster and one chain.

Reproducibility

All solvers, generators, verification scripts, datasets, and the witness plays quoted above are available in the repository accompanying this paper. Every finite claim marked machine-verified can be reproduced from a cold start in under an hour on a laptop.

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